Software Project Management

M1 — Project Initiation Report

Project title: GoodTracker - Mobile Equipment Reservation Application

Team name/number: GoodTracker

Team members and roles: Talha Yılmaz - mobile UI; Eylül Melda Çakmak - mobile integration and testing; Kerem Tuna - database and equipment API; Kardelen Darğın - reservation API and project tracking

Submission date: 21 October 2026

Document version: 1.0 - team reviewed, 21 October 2026

Repository/project board link: https://github.com/keremtuna0/good_tracker | https://github.com/users/keremtuna0/projects/2

1. Executive Summary

GoodTracker is our planned Android app for reserving shared laboratory and office equipment. Borrowers need a simple way to see what is available and request an item without creating overlapping bookings. Staff need a way to keep equipment records and decide on requests. For the first version, we will focus on sign-in, equipment browsing, availability, reservation requests and cancellation, and staff approval. Payments, university single sign-on, push notifications and an app-store release are outside this scope. In M1, we set the project objectives and boundaries, mapped the main stakeholder groups, chose a hybrid delivery approach, and listed success checks and initial risks. These are planning decisions; the app and its tests are not complete yet. The stakeholder needs are initial assumptions because no interview evidence is recorded. Our next step is to submit this report through DYS by 22 October 2026, then prepare the M2 backlog, estimates, schedule and quality plan for 1 November. We will use repository commits and the project board to track actual work.

2. Project Charter

2.1 Objectives

Objective	Success measure	Target
Find and reserve an item	Walk through sign-in, finding an item, availability, request and cancellation on Android.	The full flow works for a sample item.
Prevent double bookings	Submit two overlapping requests for one item and time slot.	At most one request is accepted.
Separate staff and user actions	Try staff actions with staff and regular user accounts.	Only staff manage items and approvals; users change only their own requests.

The targets above will be checked on a working build. They are not test results yet.

2.2 Scope

In scope	Out of scope
Sign-in, equipment browsing, search by category or location, and availability.	Payments and university single sign-on.
Reservation requests, cancellation, status, and conflict prevention.	Push notifications and an app-store release.
Staff equipment management and approval or rejection of requests.	A production rollout across campus.

The proposed stack is React Native with Expo and TypeScript, Node.js with Express, and Supabase for data and sign-in. We will check it with a small prototype before committing to it.

2.3 Constraints

The work must fit the SE3003 milestones and a four-person student team. The M1 report is due through DYS on 22 October 2026, with one team member submitting for the group. The first demo is limited to Android. We do not yet have confirmed access to real equipment records or users, and no paid-service budget is recorded. The app and backend have not been built yet.

2.4 Assumptions

ID	Assumption	If incorrect, what may happen?
A-01	Students or staff need to see availability and request shared equipment; user feedback is still needed.	The target users or flow may change.
A-02	A staff member will maintain item records and approve or reject requests.	Roles or approvals may need to change.
A-03	Expo, Express and Supabase will support a small Android demo within available limits.	We may need to change the setup or scope.
A-04	We can prepare sample equipment and reservation data.	Data preparation may take extra time.

3. Stakeholder Analysis

Stakeholder	Interest / Need	Influence	Engagement approach
Borrowers / end users	Availability and a clear request process (assumed)	High	Ask potential users to try a prototype and comment on the flow.
Inventory staff	Accurate item records and request decisions (assumed)	High	Check the proposed approval flow with a staff representative.
Development team	Clear scope, roles and decisions	High	Use issues, the board and reviews to coordinate work.
Instructor / TAs	Milestone evidence and course requirements	High	Submit the official report through the required course channel.

Stakeholder map: Manage closely - borrowers, the development team and course staff. Keep satisfied - inventory staff. Keep informed - the wider campus community. Monitor - external service providers. Needs attributed to borrowers and inventory staff are assumptions until we collect feedback.

4. Delivery Approach Decision

Predictive is too rigid for an untested booking flow; pure Agile misses the fixed M1-M5 reports. We chose Hybrid so our four-person team can plan milestones and refine the app in short cycles. We will detail these cycles in M2.

5. Initial Success Criteria

Area	Criterion	How it will be verified
Product	No overlapping confirmed reservations for one item and time slot.	Submit overlapping requests and check that at most one succeeds.
Product	Regular users cannot use staff actions or change others' requests.	Try protected actions using a regular user account and check rejection.
Product	The Android demo supports the core reservation flow.	Walk through sign-in, browsing, availability and requesting on a named build.
Project management	The demo stays within the agreed Android, API and database scope.	Compare completed backlog items with the M1 scope.
Quality	Changes have a visible review and result record.	Check commits, reviews and test notes before closing work.

These are checks for later builds. No application test results are claimed in this M1 report.

6. Initial Risk Register

ID	Risk	Probability	Impact	Response	Owner	Status
R-01	Exam and coursework load delays tasks	High	Medium	Keep tasks small and plan around exam weeks.	Kardelen	Open
R-02	Supabase limits affect the demo	Medium	High	Try the stack early and track service usage.	Kerem	Open
R-03	Two users book one slot	Medium	High	Use an atomic booking rule and test simultaneous requests.	Kardelen	Open
R-04	Android screens display poorly	Medium	Medium	Check the demo on different screen sizes.	Eylül	Open
R-05	Mobile work waits for API design	Medium	High	Agree API contracts and use sample responses early.	Talha	Open
R-06	Extra features delay the core flow	Medium	Medium	Prioritize the MVP in M2 and defer extras.	Kerem	Open

7. Planned vs. Actual

This comparison uses the assigned M1 outputs and repository history. We do not have a dated effort baseline, so it would be misleading to claim a schedule variance in hours or days.

Planned activity	Actual result	Deviation	Reason	Corrective action
Project charter	Charter added to main (a3496ca) and reviewed by the team.	No remaining review action	The section was drafted first.	No further action from the team review.
Stakeholder analysis	Talha added the stakeholder section (9301b96).	Needs unverified	No interview evidence is recorded.	Seek feedback on a prototype.
Success criteria	Eylül added criteria (4d9ba27).	Tests not run	No app build exists yet.	Run checks on a named build.
Delivery approach and risks	Kardelen added the section; PR #26 was merged.	Review record absent	No peer review is recorded on the PR.	Record peer reviews on future PRs.
M1 report	Word report completed and reviewed; DYS deadline confirmed.	DYS submission pending	DYS upload has not happened yet.	Submit once through DYS by 22 October.

8. Next Milestone Plan

Item	Owner	Target date	Expected evidence
Prioritized backlog and acceptance criteria	Talha	1 November 2026 (DYS)	Backlog linked to issues (#6).
Effort estimates and assumptions	Kerem	1 November 2026 (DYS)	Estimates and assumption log (#7).
Schedule, RACI and communication plan	Kardelen	1 November 2026 (DYS)	Schedule and team documents (#8).
Quality plan and baseline report	Eylül	1 November 2026 (DYS)	Quality plan and M2 report (#9).

The weekly report for 19-25 October is a separate submission due through DYS on 26 October 2026 at 23:59.

9. Individual Contributions

Team member	Contribution	Evidence
Talha Yılmaz	Stakeholder analysis and map.	stakeholders.md; commit 9301b96
Eylül Melda Çakmak	Initial success criteria and planned checks.	success-criteria.md; commit 4d9ba27
Kardelen Dargın	Delivery approach and initial risk register.	approach-and-risks.md; commit f9e9e7e; PR #26
Kerem Tuna	Project charter, risk wording alignment and coordination of this report.	charter.md; commits a3496ca and 6851aab; this report

Appendix: Evidence

The links below show the work available in the repository and the DYS deadline announcement. The team reviewed this report. Interview notes, executed tests and a DYS submission receipt are not available yet.

Repository and project documents - current project source and M1 files.

Project board - task workflow and future progress tracking.

Charter commit - objectives, scope, constraints and assumptions.

Stakeholder commit - stakeholder analysis by Talha.

Success criteria commit - initial checks by Eylul.

Delivery approach PR - approach and risk section by Kardelen, merged into main.

DYS deadline announcement - shows M1 due 22 October, M2 due 1 November, and the separate weekly report due 26 October 2026 at 23:59.